

Daniel Lee

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EDUCATION

University of Michigan

Bachelor of Science in Computer Science, Minor in Mathematics

Ann Arbor, MI

Expected May 2028

- **Relevant Coursework:** Web Systems (in progress), Data Structures and Algorithms, Computer Organization, Discrete Mathematics, Linear Algebra, Introduction to Data Science

TECHNICAL SKILLS

Languages: C++, Python, Java, JavaScript, TypeScript, SQL, Swift, HTML/CSS

Frameworks: React, React Native, Node.js, Express, Flask, Spring Boot, SwiftUI

Tools & Data: Git, Linux, CMake, PostgreSQL, Supabase, OpenAI API, pandas, NumPy

EXPERIENCE

Undergraduate Research Assistant

May 2026 – Present

University of Michigan, PROTEUS Project (NSF-Funded)

Ann Arbor, MI

- Built a React/TypeScript dashboard (Vite, Tailwind) used by 7 researchers to explore 80K+ student submissions, replacing manual spreadsheet analysis
- Engineered a Python/pandas ETL pipeline that ingests, validates, and normalizes 960K+ log events into 80K+ submissions across 10 course sections and 2 textbooks
- Diagnosed a silent data-loss bug where 512-character field truncation dropped ~30% of long submissions; rerouted parsing to the untruncated field, correcting a key accuracy metric from 16.3% to 7.1%

Project Team Member

Sep. 2024 – Dec. 2024

Michigan Data Science Team

Ann Arbor, MI

- Built Python/pandas pipelines to clean and analyze 7,000+ criminal records from the COMPAS recidivism dataset
- Identified significant racial disparities in COMPAS risk scores using logistic regression and survival analysis; presented findings to 100+ attendees at the MDST Project Expo

PROJECTS

Limit Order Book | C++, CMake | [GitHub](#)

June 2026

- Implemented a price-time priority limit order book in C++, using sorted `std::map` price levels for bids/asks and a hash index for O(1) lookup
- Replayed a real Nasdaq ITCH market-by-order feed of 4.88M events at a median 12.8M events/sec (~78 ns/event), maintaining live book state and bid-ask spread

LeetQuest | React, TypeScript, Flask, Python, OpenAI API | [GitHub](#)

DivHacks, Oct. 2025

- Built the Flask backend for a coding practice web app (team of 3), serving REST endpoints that filter 500+ LeetCode problems by topic and difficulty into tailored problem sets
- Integrated the OpenAI API to power an AI tutor that gives hints, code explanations, and debugging help without revealing full solutions

TrainSense | React Native, Flask, Python, pandas | [GitHub](#) | [Demo](#)

June 2025

- Built a full-stack subway trip planner with a React Native client and a Flask REST backend
- Modeled the subway network as a graph from GTFS data and implemented BFS pathfinding to compute station-to-station routes

Spotlight | React, Vite, Tailwind CSS, Mapbox GL JS, Supabase, Node.js | [GitHub](#) | [Demo](#)

May 2025

- Built a full-stack event-discovery app aggregating live listings from the Ticketmaster and SeatGeek APIs into a Supabase/PostgreSQL backend
- De-duplicated cross-source listings using token-overlap name matching and venue distance, handling events listed under different names
- Implemented map clustering, photo markers, and fuzzy search with Mapbox GL JS

LEADERSHIP AND PROFESSIONAL DEVELOPMENT

Career Prep Fellow

Jan. 2026 – Present

Management Leadership for Tomorrow

- Selected for an 18-month career development program providing coaching and mentorship to high-potential students